SOP-114

Boiler Feed Pump — Low Suction Pressure Response

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1 Purpose and Scope

This procedure defines the required operator response to a Suction Pressure Low alarm on any boiler feed pump at NorthPlant. It applies to Boiler Feed Pump 101 and Boiler Feed Pump 102 in Unit 1, and to equivalent duty pumps installed elsewhere on site.

The procedure covers immediate response, diagnosis of the suction-side cause, and the criteria for escalation to Maintenance. It does not cover mechanical repair, which is addressed by MM-207.

NOTE Suction Pressure Low is classified Priority 2 (High) under AP-001. The target operator response time for a Priority 2 alarm is 10 minutes.

2 Safety Precautions

All work under this procedure is performed from the control room or from outside the pump guard line. No part of this procedure authorises physical contact with pump internals, couplings or the drive train.

WARNING Any inspection requiring access to rotating parts must follow SI-009. Isolation, lockout and stored-energy dissipation are mandatory and are not waived by a diagnostic recommendation from any monitoring or advisory system.

3 Immediate Operator Response

3.1 Acknowledgement

Acknowledge the alarm within the Priority 2 response target. Record the acknowledgement time; sustained acknowledgement delay above 15 minutes is itself reportable under AP-001 §5.2.

3.2 Suction-Side Checks

On a Suction Pressure Low alarm, the operator shall first confirm that deaerator level is above the 60 percent low-low setpoint, and verify that the suction strainer differential pressure is below 0.4 bar. Where the differential exceeds 0.4 bar, initiate strainer changeover before reducing pump demand.

1. Confirm deaerator level is above the 60 percent low-low setpoint.
2. Read suction strainer differential pressure at the local indicator or DCS tag.
3. If differential pressure exceeds 0.4 bar, initiate strainer changeover to the standby element and log the changeover.
4. If differential pressure is below 0.4 bar, proceed to §3.3.
5. Confirm the suction valve is fully open and its position feedback agrees with the commanded position.

> **NOTE** A strainer changeover that clears the alarm within five minutes indicates fouling rather than a system-level NPSH deficit. Record the changeover so that recurrence can be trended.

3.3 Demand Reduction

If suction pressure has not recovered after the checks in §3.2, reduce pump demand in steps of 10 percent, allowing two minutes at each step for the suction condition to stabilise. Do not reduce below the minimum flow setpoint; operation below minimum flow introduces recirculation damage that is more costly than the condition being avoided.

> **WARNING** Do not attempt to clear a persistent Suction Pressure Low alarm by raising the alarm setpoint. Setpoint changes require rationalisation under AP-001 §6.

4 Diagnosis of Recurring Events

A single Suction Pressure Low event that clears on strainer changeover is treated as a routine fouling event. Recurring events indicate a systemic cause and shall be investigated rather than repeatedly ridden through.

Occurrences in 30 days	Classification	Required action
1 to 2	Routine	Log and trend. No escalation.
3 to 5	Recurring	Raise an investigation. Recalculate available NPSH.
More than 5	Chronic	Escalate to Maintenance under MM-207 §7.3.

Table 4-1 — Recurrence classification

Where Suction Pressure Low events are accompanied by a rise in bearing temperature within 10 to 20 minutes, treat the pattern as suspected intermittent cavitation and refer to TG-051 §2.4 before scheduling any mechanical intervention.

4.1 Evidence to Collect

- Alarm count and recurrence rate for the preceding 90 days.
- Median lag between the suction alarm and any associated bearing temperature alarm.
- Strainer differential pressure trend across the same period.
- Deaerator level trend, including any excursions below the low-low setpoint.
- Record of strainer changeovers and their effect on the alarm.

5 Escalation Criteria

Escalate to Maintenance where any of the following applies:

- The condition is classified Chronic under Table 4-1.
- Bearing temperature does not return to its normal band within 30 minutes of suction pressure recovering.
- Available NPSH calculated at current duty is within 0.5 m of required NPSH.
- A seal leak is detected at any time during the event.

Escalation is by maintenance work request. Operators do not schedule mechanical work directly; the inspection interval is set by MM-207 §7.3 and, where that section and a monitoring recommendation disagree, MM-207 governs.

6 Records

Record the alarm identifier, acknowledgement time, checks performed, whether a strainer changeover was initiated, and the outcome. Retain for 24 months.